Lambda 1 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 5 complete ICs for all displayed variants

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
1 ANN10	**base10k**	0.15	0.476	1.02	1.04	0.329	5.49	0.509	0.967	0.997	0	18/18
1 ANN10	**soft10k**	0.352	0.89	14.1	0.681	0.468	1.12	0.646	3.92	2.51	0	17/18
1 ANN10	**long**	0.286	0.426	18.7	0.728	0.462	2.03	0.629	2.11	1.49	0	18/18
2 P-only DeepONet10	**base10k**	8.74e-03	0.236	1	1	0.0395	1	1	0.997	1	0	18/18
2 P-only DeepONet10	**soft10k**	0.375	0.98	1.42	0.93	0.534	0.732	0.664	1.02	3.55	0	18/18
2 P-only DeepONet10	**long**	0.371	0.957	2.1	1.18	0.552	0.838	0.66	1.28	1.08	0	18/18
3 ANN10 + P-only DeepONet10	**base10k**	9.87e-03	0.233	1	1	0.0434	0.998	0.999	0.988	0.999	0	18/18
3 ANN10 + P-only DeepONet10	**soft10k**	0.376	0.982	1.51	0.703	0.533	0.668	0.666	1.98	0.997	0	18/18
3 ANN10 + P-only DeepONet10	**long**	0.373	0.979	1.65	0.956	0.527	0.77	0.663	1.08	1.01	0	18/18
4 T/P DeepONet10	**base10k**	0.0406	0.212	1	0.887	0.0858	0.952	0.626	0.989	0.992	0	17/18
4 T/P DeepONet10	**soft10k**	0.368	0.906	4.26	1.14	0.485	1.28	0.671	4.83	7.85	0	9/18
4 T/P DeepONet10	**long**	0.479	0.982	6.25	1.98	0.554	1.05	0.99	103	487	0	14/18
5 ANN10 + T/P DeepONet10	**base10k**	0.0362	0.13	1	0.728	0.0804	0.999	0.606	0.921	0.99	0	18/18
5 ANN10 + T/P DeepONet10	**soft10k**	0.374	0.956	2.56	0.739	0.506	0.71	0.663	0.884	0.998	0	18/18
5 ANN10 + T/P DeepONet10	**long**	0.172	0.186	8.61	3.08	0.321	3.7	0.635	0.898	0.996	0	18/18
6 T/P/ϕ DeepONet10	**base10k**	0.0714	0.197	1	1	0.196	1	1	0.999	1	0	17/18
6 T/P/ϕ DeepONet10	**soft10k**	0.624	0.971	9.46	0.793	0.512	1.49	0.676	0.995	1	0	11/18
6 T/P/ϕ DeepONet10	**long**	0.352	0.968	20.4	4.07	0.527	5.15	0.624	1.94	0.993	0	13/18
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	0.0351	0.121	1	1	0.0652	1	1	0.998	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.382	0.978	2.89	0.907	0.519	0.98	0.674	0.894	0.999	0	17/18
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.359	0.916	10.2	2.11	0.485	3.78	0.641	1.2	0.994	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 39, 64, 69.